

Introduction to Deep Equilibrium Nets (DEQNs)

A Mini-Course

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DOI: <https://doi.org/10.1111/iere.12575>

Materials: https://github.com/sischei/DEQN_Galo_03_2026

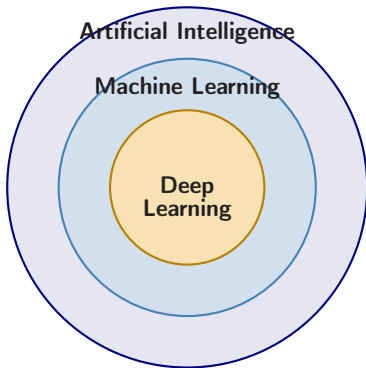
Course Outline

1. ML/DL Foundations
2. Deep Equilibrium Nets
3. Brock–Mirman Benchmark
4. Practical Tools: NAS & Loss Normalization
5. Scaling Up — The IRBC Model
6. Deep Surrogate Models
7. Hands-on Exercises

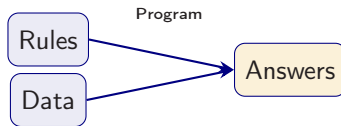
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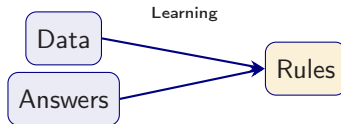
What is Machine Learning?



Classical programming:

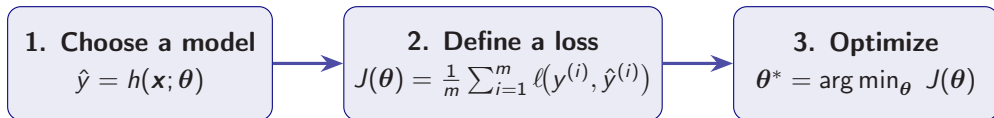


Machine learning:



The Machine Learning Recipe

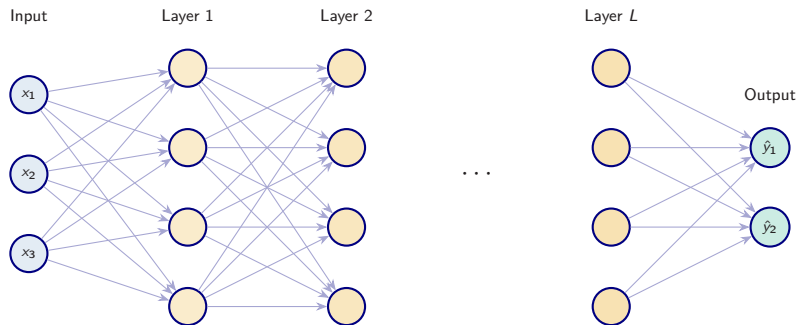
Every supervised learning algorithm follows three steps:



Cost function example (Mean Squared Error)

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)})^2$$

Deep Feedforward Networks



An L -layer network computes a nested composition:

$$f(\mathbf{x}; \boldsymbol{\theta}) = g_L\left(\mathbf{W}^{(L)} g_{L-1}(\cdots g_1(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)}) \cdots + \mathbf{b}^{(L-1)}) + \mathbf{b}^{(L)}\right)$$

Each layer transforms the representation. **Depth** provides exponential gains in representational power over width alone.

Universal Approximation Theorem

Theorem (Cybenko 1989; Hornik, Stinchcombe & White 1989)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded, non-constant, continuous activation function. Then for any continuous function $f : \mathcal{K} \rightarrow \mathbb{R}$ on a compact set $\mathcal{K} \subset \mathbb{R}^d$ and any $\varepsilon > 0$, there exists a single-hidden-layer network

$$F(\mathbf{x}) = \sum_{j=1}^{n_1} v_j g(\mathbf{w}_j^\top \mathbf{x} + b_j)$$

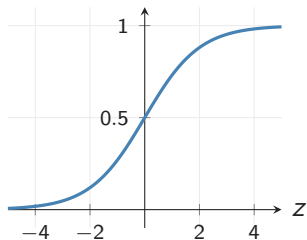
such that $\sup_{\mathbf{x} \in \mathcal{K}} |F(\mathbf{x}) - f(\mathbf{x})| < \varepsilon$.

Caveats:

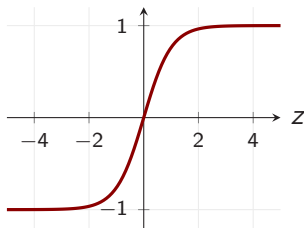
- ▶ The required width n_1 may be **exponentially large** in d .
- ▶ Existence $\neq$ efficient learnability.
- ▶ Practical success of deep learning suggests that **depth** provides a much more efficient parameterization than width alone.

Activation Functions

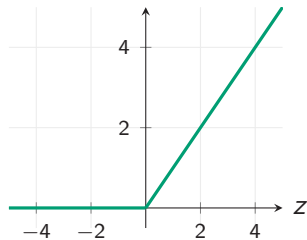
Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$



Tanh: $\tanh(z)$

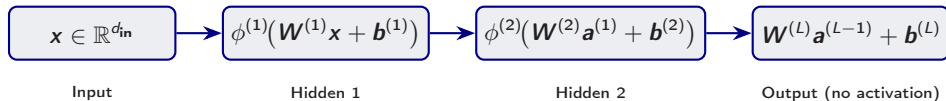


ReLU: $\max(0, z)$



Other common activations: **Leaky ReLU** $\max(0.01z, z)$, **Softplus** $\ln(1 + e^z)$, **ELU**, **Swish** $z \cdot \sigma(z)$.

DNN Layer Structure



Each hidden layer applies a **nonlinear activation** $\phi^{(\ell)}$:

$$\mathbf{a}^{(\ell)} = \phi^{(\ell)}(\mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}), \quad \ell = 1, \dots, L-1$$

- ▶ The output layer is typically **linear** (for regression) or uses **softplus** (to enforce positivity, e.g. consumption)
- ▶ The full network: $\mathcal{N}_{\rho}(\mathbf{x}) = \mathbf{W}^{(L)}\mathbf{a}^{(L-1)} + \mathbf{b}^{(L)}$

Gradient Descent and SGD

Goal: find weights that minimize the loss:

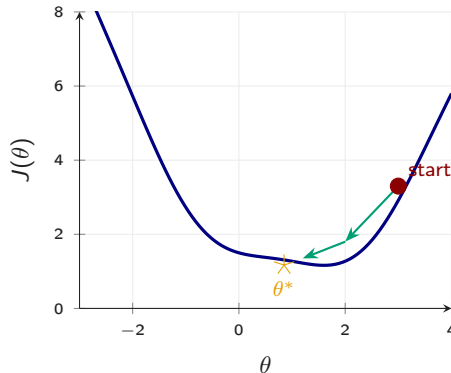
$$\theta^* = \arg \min_{\theta} J(\theta)$$

Strategy: repeatedly step in the direction of steepest descent:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} J(\theta)$$

Mini-batch SGD: sample a batch $\mathcal{B}$ of B examples:

$$\theta \leftarrow \theta - \frac{\eta}{B} \sum_{k \in \mathcal{B}} \nabla_{\theta} J_k(\theta)$$



Loss Functions

The loss function $J(\theta)$ measures how far predictions $\hat{y}$ are from targets y .

Regression: Mean Squared Error

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n (y^{(i)} - f(\mathbf{x}^{(i)}; \theta))^2$$

Penalizes large deviations quadratically.

Classification: Cross-Entropy

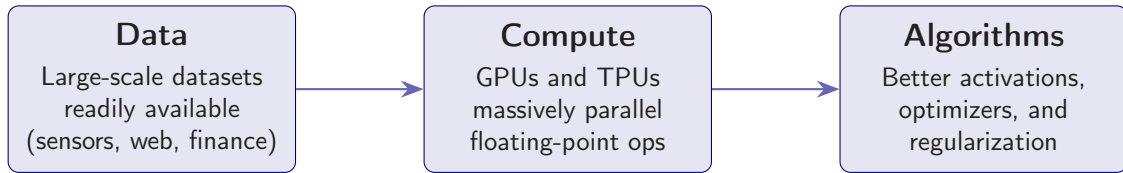
$$J = -\frac{1}{n} \sum_{i=1}^n [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log (1 - \hat{y}^{(i)})]$$

For binary labels $y \in \{0, 1\}$.

In computational economics, we will replace labeled data with **structural equations**—the key insight behind DEQNs.

Why Deep Learning Now?

Neural networks date back to the 1940s. Three developments made them practical:



Key milestones: Perceptron (1958), Backpropagation (1986), Deep CNN (1995), AlexNet (2012), Transformers (2017), ChatGPT (2022).

Outline

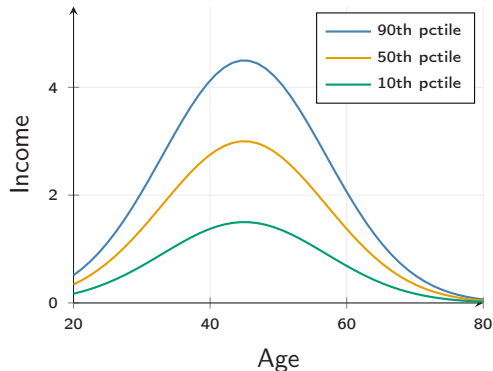
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Heterogeneity Matters

Key questions in macroeconomics:

- ▶ How does **inequality** evolve over the life cycle?
- ▶ How do **idiosyncratic shocks** interact with aggregate risk?
- ▶ What are the **distributional effects** of policy?

⇒ Need **overlapping-generations** (OLG) models with many heterogeneous agents.



Stylized income-by-age profiles

Dynamic Stochastic General Equilibrium Models

Generic formulation: find policy functions $p(\cdot)$ such that

$$\mathbb{E}\left[G(\mathbf{x}_t, \mathbf{x}_{t+1}, p(\mathbf{x}_t), p(\mathbf{x}_{t+1})) \mid \mathbf{x}_t, p(\mathbf{x}_t)\right] = 0$$

with transition law $\mathbf{x}_{t+1} = h(\mathbf{x}_t, p(\mathbf{x}_t), \varepsilon_{t+1})$.

Challenges:

- ▶ State space $\mathbf{x}_t \in \mathbb{R}^d$ can be **very high-dimensional**
- ▶ Standard grid-based methods scale as $\mathcal{O}(n^d) \Rightarrow$ **curse of dimensionality**
- ▶ Need a method that **does not rely on grids**

What is High-Dimensional?

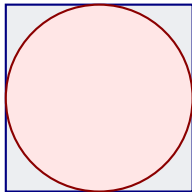
Dimensions d	Grid points n^d	Runtime
1	10	10 seconds
2	100	2 minutes
5	10^5	1 day
10	10^{10}	317 years
20	10^{20}	~ 3 trillion years

Assumes $n = 10$ grid points per dimension and 1 second per evaluation.

Three remedies:

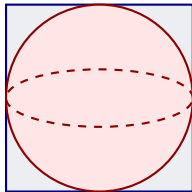
1. **Sparse grids** (Brumm & Scheidegger, 2017)
2. **Random grids** (Rust, 1997)
3. **Neural networks** (this lecture)

Volumes in High Dimensions



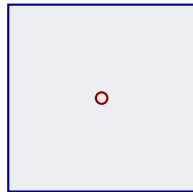
2D

$$V_{\text{sphere}}/V_{\text{cube}} = \pi/4 \approx 0.785$$



3D

$$V_{\text{sphere}}/V_{\text{cube}} = \pi/6 \approx 0.524$$



100D

$$V_{\text{sphere}}/V_{\text{cube}} \approx 1.87 \times 10^{-70}$$

Implication

In high dimensions, **almost all volume is in the corners**. Grid-based methods waste most of their evaluation points in regions that barely contribute to the integral or approximation.

Our Solution: Deep Equilibrium Nets

Core idea (Azinovic, Gaegauf & Scheidegger, 2022):

Replace the traditional grid-based solution with a **neural network** trained to satisfy economic equilibrium conditions.

Three Key Ingredients

1. **Equilibrium error as loss function:** use the economic equations $G(\cdot) = 0$ directly—**no labeled data** needed
2. **Stochastic gradient descent:** optimize network parameters ρ via SGD on mini-batches
3. **Simulated paths:** draw training points from the **ergodic distribution** of the model itself

⇒ This turns **solving** an economic model into **training** a neural network.

What is a Deep Neural Network?

A deep neural network (DNN) $\mathcal{N}_\rho : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}$ is a **parametric function approximator**.

Universal approximation (Cybenko 1989, Hornik 1991):

For any continuous $f : \mathcal{K} \rightarrow \mathbb{R}$ and $\varepsilon > 0$, $\exists \rho^\star : \quad \|\mathcal{N}_{\rho^\star} - f\| < \varepsilon$

In economics:

- ▶ **Input:** state variables $\mathbf{x}_t \in \mathbb{R}^d$ (capital, productivity, wealth distribution, ...)
- ▶ **Output:** policy variables (consumption, savings, prices, ...)
- ▶ **Parameters** ρ : weights and biases—found by **training**

The “Data Problem” in Economics

Deep learning excels when massive labeled datasets are available.

But in computational economics:

- ▶ We solve **functional equations**, not classification/regression tasks
- ▶ There are **no labeled pairs** $(\mathbf{x}_i, y_i)$
- ▶ Traditional approach: **time iteration on a grid**—but grids do not scale

Key insight:

- ▶ We **know** the structural equations $G(\cdot) = 0$
- ▶ We can **check** how well a candidate policy satisfies them
- ▶ $\Rightarrow$ Use the **equilibrium error** as the loss function!

Traditional Approach: Time Iteration on a Grid

Algorithm: Grid-Based Collocation

Input: Grid $\mathcal{G} = \{\mathbf{x}_1, \dots, \mathbf{x}_M\} \subset \mathbb{R}^d$, initial guess $p^{(0)}$

for $k = 0, 1, 2, \dots$ until convergence **do**

for each grid point $\mathbf{x}_i \in \mathcal{G}$ **do**

 Solve $G(\mathbf{x}_i, p^{(k)}(\mathbf{x}_i), p^{(k)}) = 0$ for $p^{(k+1)}(\mathbf{x}_i)$

end for

 Interpolate $p^{(k+1)}$ between grid points

end for

Problems:

- ▶ Grid size $M = n^d \Rightarrow$ **exponential** in dimension d
- ▶ Interpolation in $d > 5$ becomes expensive and inaccurate
- ▶ Cannot handle models with $d = 60$ or more state variables

The DEQN Loss Function

Key idea: replace labeled data with **economic equilibrium conditions**.

Let $G(\mathbf{x}, f(\mathbf{x})) = 0$ be the system of equilibrium equations. Define:

$$\ell_{\rho} = \frac{1}{N} \sum_{i=1}^N \|G(\mathbf{x}_i, \mathcal{N}_{\rho}(\mathbf{x}_i))\|^2$$

- ▶ $\mathcal{N}_{\rho}(\mathbf{x}_i)$ is the neural net's **predicted policy** at state $\mathbf{x}_i$
- ▶ $G(\cdot)$ measures the **equilibrium error** (e.g. Euler equation residual)
- ▶ $\ell_{\rho} = 0$ iff the policy **exactly** satisfies all equilibrium conditions

⇒ **No labels needed**—this is **unsupervised** (or “self-supervised”) learning.

Training Algorithm for DEQNs

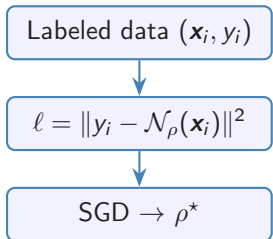
Algorithm 1: Deep Equilibrium Net Training

Input: Initial state $\mathbf{x}_0$, network $\mathcal{N}_\rho$, learning rate α , # episodes E , # epochs T
for episode $e = 1, \dots, E$ **do**
 Simulate path: $\mathbf{x}_0 \rightarrow \mathbf{x}_1 \rightarrow \dots \rightarrow \mathbf{x}_T$ using $\mathcal{N}_\rho$ and transition law
 for epoch $t = 1, \dots, T$ **do**
 Draw mini-batch $\mathcal{B} \subset \{\mathbf{x}_0, \dots, \mathbf{x}_T\}$
 Compute loss: $\ell_\rho = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} \|G(\mathbf{x}_i, \mathcal{N}_\rho(\mathbf{x}_i))\|^2$
 Update: $\rho \leftarrow \rho - \alpha \cdot \nabla_\rho \ell_\rho$
 end for
end for
Output: Trained network $\mathcal{N}_{\rho^*}$ approximating the policy function

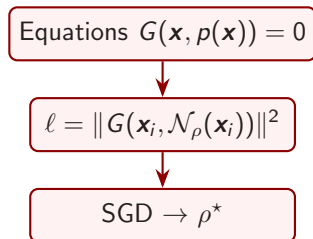
Note: Training points come from the **model's own simulation**—the network learns on the ergodic distribution, focusing on economically relevant regions.

Supervised vs. Unsupervised (DEQN) Learning

Supervised



DEQN (Unsupervised)



Key difference: DEQNs replace labeled data with structural economic equations.

The Cost of Labeled Data in Economics

Supervised learning requires labeled pairs $(\mathbf{x}_i, y_i)$.

In computational economics, generating a single label y_i means **solving the model** at state $\mathbf{x}_i$ —precisely the problem we are trying to solve!

Step	Supervised (label generation)	DEQN (unsupervised)
1	Choose grid $\mathcal{G}$ of n^d points	Draw states from simulation
2	Solve $G(\mathbf{x}_i, y_i) = 0$ at each $\mathbf{x}_i$	Evaluate residual $G(\mathbf{x}_i, \mathcal{N}_\rho(\mathbf{x}_i))$
3	Fit $\hat{y} = f(\mathbf{x}; \theta)$ to $\{(\mathbf{x}_i, y_i)\}$	Update ρ via SGD on residual
Cost	$\underbrace{n^d}_{\text{grid}} \times \underbrace{C_{\text{solve}}}_{\text{per point}} + \text{fitting}$	$E \times T \times \underbrace{B}_{\text{batch}} \times C_{\text{residual}}$

- ▶ C_{solve} : cost of solving a nonlinear system or iterating a Bellman equation to convergence
- ▶ C_{residual} : cost of a single forward pass through the equilibrium equations—**orders of magnitude cheaper**

Why Supervised Learning is Prohibitively Expensive

To generate labels, we must solve the model on a grid. This involves:

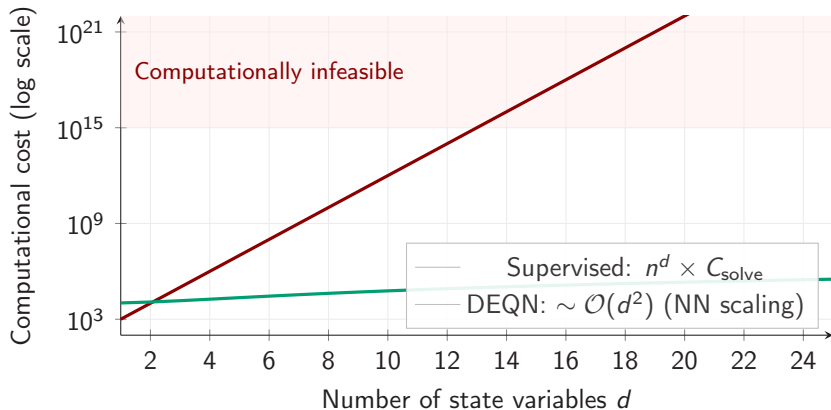
1. Bellman / value function iteration

- ▶ Iterate
$$V_{k+1}(\mathbf{x}) = \max_a \{u(a) + \beta \mathbb{E}[V_k(\mathbf{x}')]\}$$
- ▶ Each iteration: solve n^d optimisation problems
- ▶ Typically $\mathcal{O}(100)$ iterations to converge
- ▶ Interpolation between grid points introduces error

2. Nonlinear system solving

- ▶ At each $\mathbf{x}_i$: solve $G(\mathbf{x}_i, y) = 0$ via Newton's method
- ▶ Each Newton step: compute and invert a Jacobian
- ▶ Cost per point: $\mathcal{O}(m^3)$ for m equations
- ▶ Convergence not guaranteed away from the solution

Computational Cost: Supervised vs. DEQN



Supervised approaches hit the wall around $d \approx 5-8$. DEQNs scale gracefully to $d > 100$.

DEQN: Key Recap

Replace Labels

Instead of labeled data $(\mathbf{x}_i, y_i)$, use the economic equilibrium equations $G(\cdot) = 0$ as the loss.

Replace Grids

Instead of n^d grid points, use a neural network $\mathcal{N}_\rho$ that maps states to policies **globally**.

Replace TI

Instead of time iteration with interpolation, use SGD on simulated paths from the model.

Result: A method that scales to **hundreds of state variables** and runs on GPUs.

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Brock–Mirman (1972): Setup

Social planner's problem:

$$\max_{\{C_t\}_{t=0}^{\infty}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \ln(C_t) \right]$$

subject to:

$$K_{t+1} + C_t = z_t K_t^{\alpha}, \quad \ln z_{t+1} = \varrho \ln z_t + \sigma_z \epsilon_{t+1}, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

Euler equation:

$$\frac{1}{C_t} = \beta \mathbb{E} \left[\frac{\alpha z_{t+1} K_{t+1}^{\alpha-1}}{C_{t+1}} \right]$$

Analytical solution (log utility, full depreciation $\delta = 1$):

$$K_{t+1} = \beta \alpha K_t^{\alpha}, \quad C_t = (1 - \beta \alpha) K_t^{\alpha}$$

DEQN for Brock–Mirman

Step 1: Parametrize the policy function with a neural network:

$$C_t = \mathcal{N}_\rho(K_t, z_t) \quad (\text{consumption as a function of state})$$

Step 2: Define the equilibrium residual:

$$G(K_t, z_t) = 1 - \beta \frac{C_t}{C_{t+1}} \cdot \alpha z_{t+1} K_{t+1}^{\alpha-1}$$

where $K_{t+1} = z_t K_t^\alpha - \mathcal{N}_\rho(K_t, z_t)$ and $C_{t+1} = \mathcal{N}_\rho(K_{t+1}, z_{t+1})$.

Step 3: Minimize the DEQN loss:

$$\ell_\rho = \frac{1}{N} \sum_{i=1}^N |G(K_t^{(i)}, z_t^{(i)})|^2$$

We can **verify** the solution against the analytical benchmark.

Hands-on Notebooks

Two notebooks accompany this lecture:

1. `01_Brock_Mirman_1972_DEQN.ipynb`
Deterministic Brock–Mirman with analytical benchmark.
2. `02_Brock_Mirman_Uncertainty_DEQN.ipynb`
Stochastic version with TFP shocks ($\varrho > 0$, $\sigma_z > 0$).

All notebooks are in the `code/` directory and on the course GitHub repository.

Note: Exercises (`03_DEQN_Exercises_Blancs.ipynb`) build on these notebooks.

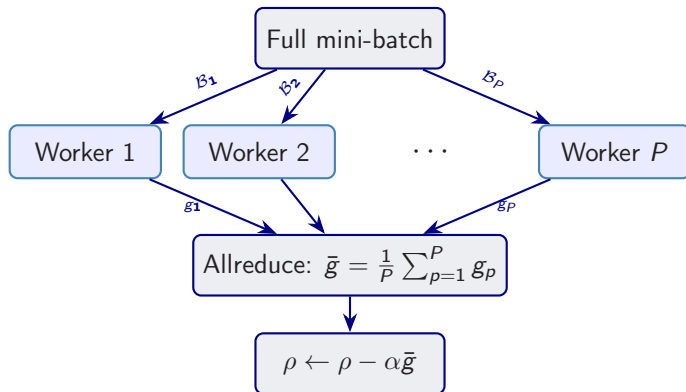
*“To pull a bigger wagon, it is easier to add more oxen
than to grow a gigantic ox.”*

— Gropp, Lusk & Skjellum (1999)

Using MPI: Portable Parallel Programming

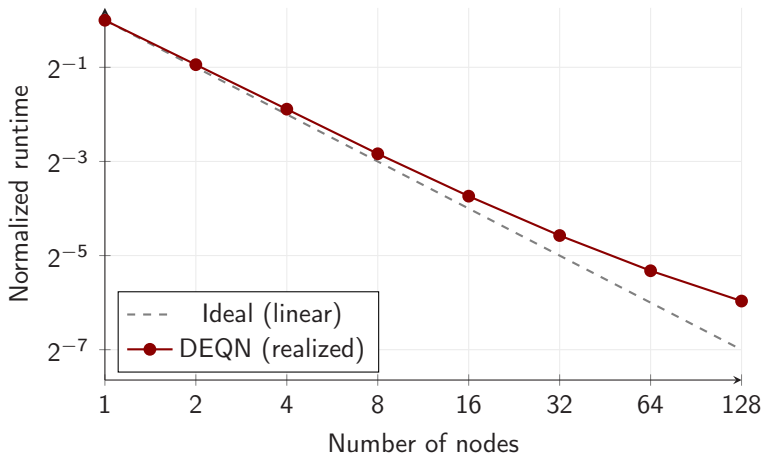
Data Parallelism for DEQNs

Idea: distribute training data across **multiple workers** (GPUs/CPUs).



Implemented via **Horovod**/MPI. Each worker computes a local gradient; **allreduce** averages them.

Scalability Results



Near-linear scaling up to 128 nodes. Communication overhead is negligible for typical network sizes.

Outline

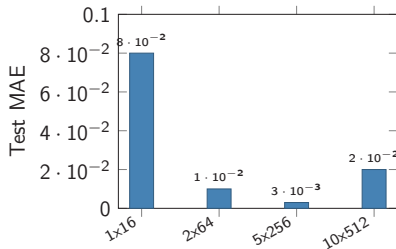
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Why Architecture Matters

Same data, different architectures:

- ▶ 1 layer, 16 units $\Rightarrow$ MAE = 0.08
- ▶ 2 layers, 64 units $\Rightarrow$ MAE = 0.01
- ▶ 5 layers, 256 units $\Rightarrow$ MAE = 0.003
- ▶ 10 layers, 512 units $\Rightarrow$ MAE = 0.02 (overfit)

Architecture choice can matter **more than training duration**.



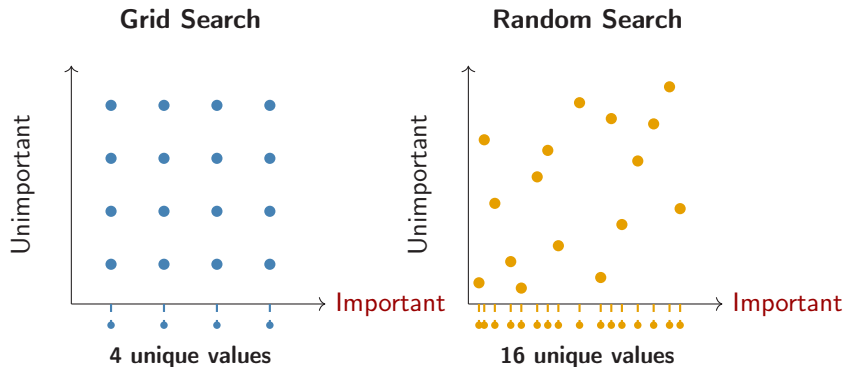
Neural Architecture Search: Method Comparison

Property	Grid	Random	Bayesian	Hyperband
Computational cost	$\mathcal{O}(k^d)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n \log n)$
Sample efficiency	Low	Medium	High	High
Parallelisable	Fully	Fully	Limited	Partially
Handles interactions	No	Partially	Yes	No
Early stopping	No	No	No	Yes
Implementation	Trivial	Trivial	Moderate	Moderate

Practical advice:

- ▶ **Quick exploration:** Random search (strong baseline)
- ▶ **Limited budget:** Bayesian optimisation (sample-efficient)
- ▶ **Large search space:** Hyperband (aggressive early stopping)

Grid vs. Random — Why Random Wins



When only one dimension matters, grid wastes $\frac{3}{4}$ of its budget on the *unimportant* axis. Random explores the important axis **4×** more densely.

Bayesian Optimisation

Idea: Build a *surrogate model* of $f(\text{HP}) \rightarrow \text{loss}$ and use it to choose the next trial *intelligently*.

1. Fit a Gaussian process (GP) to observed trials
2. Compute an **acquisition function** (e.g., Expected Improvement)
3. Evaluate the HP with highest acquisition value
4. Update the GP and repeat

Key advantage: **Sample-efficient** — exploits structure in the response surface.

Expected Improvement

$$\text{EI}(\mathbf{x}) = \mathbb{E}[\max(f^* - f(\mathbf{x}), 0)]$$

- ▶ f^* : best observed value
- ▶ Balances *exploration* (high uncertainty) and *exploitation* (low predicted loss)

NAS for Computational Economics

Where NAS matters in computational economics:

1. **Surrogate models:** Replace expensive simulations (e.g., climate IAMs) with NNs — architecture determines approximation quality.
2. **DSGE policy functions:** Deep equilibrium nets use NNs to approximate policy/value functions. **Architecture search over the policy network** can improve convergence.
3. **High-dimensional problems:** As d grows (many state variables), the architecture must scale. NAS automates this adaptation.
4. **Reproducibility:** Automated search documents *which* architectures were tried, making results more transparent than “we chose 3 layers with 64 units.”

NAS Hands-on Notebook

Hands-on: 05_Neural_Architecture_Search.ipynb

1. Define a test function (Genz Gaussian on $[0, 1]^2$)
2. Specify a hyperparameter search space:
 - ▶ Depth (1–5 layers), width (32–256), activation (ReLU, tanh, ELU, Swish)
 - ▶ Dropout, batch normalization, optimizer, learning rate, batch size
3. Run **30 random trials** with early stopping
4. Select **top-5 architectures**, retrain for longer evaluation
5. Analyse which hyperparameters matter most

Runtime: ~2–3 minutes on CPU. All code in the code/ directory.

Multi-Component Losses in DEQNs

Many problems require minimising *several* loss terms simultaneously:

Deep equilibrium nets (DEQNs):

- ▶ Euler equation residuals
- ▶ Budget constraints
- ▶ Market clearing

Physics-informed NNs (PINNs):

- ▶ PDE residual $\mathcal{L}_{\text{PDE}}$
- ▶ Boundary conditions $\mathcal{L}_{\text{BC}}$
- ▶ Initial conditions $\mathcal{L}_{\text{IC}}$

Core challenge

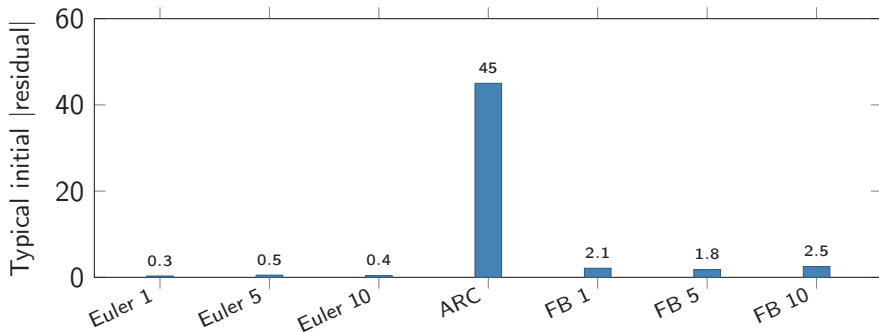
These loss components often live at **vastly different scales**.

With equal weights:

- ▶ Large-scale component dominates gradients
- ▶ Small-scale component is **ignored**
- ▶ The network effectively solves a *single-task* problem

The Scale Mismatch Problem

Example: IRBC model with $N = 10$ countries has three types of residuals:



- ▶ The **aggregate resource constraint** (ARC) is $\sim 100\times$ larger than the Euler residuals
- ▶ With equal weighting: $\nabla_{\rho} \ell \approx \nabla_{\rho} \ell_{\text{ARC}}$ —the network **only** learns to clear markets
- ▶ Euler equations and complementarity conditions are **effectively ignored**

Gradient Dominance: Why Naïve Weighting Fails

Total loss: $\ell = \sum_{k=1}^K \ell_k$. The gradient: $\nabla_{\rho} \ell = \sum_{k=1}^K \nabla_{\rho} \ell_k$.

When $\ell_j \gg \ell_k$ for some j :

- ▶ $\|\nabla_{\rho} \ell_j\| \gg \|\nabla_{\rho} \ell_k\| \Rightarrow$ update direction dominated by component j
- ▶ Components $k \neq j$ may even **increase** during training
- ▶ Network satisfies *one* equation well but violates the others

Naïve: $w_k = 1$ for all k

- ▶ Large-scale loss dominates
- ▶ Small losses stagnate or worsen
- ▶ Effectively single-task learning

Adaptive: w_k adjusted dynamically

- ▶ All components contribute equally
- ▶ Network balances all conditions
- ▶ True multi-task learning

Approaches to Loss Balancing

Method	Idea	Limitation
Equal weights	$w_k = 1/K$	Ignores scale differences
Manual tuning	Hand-pick w_k	Does not adapt during training
Inverse-loss	$w_k \propto 1/\ell_k$	Oscillates on noisy components
GradNorm (Chen+ 2018)	Match gradient norms across tasks	Per-task gradient computation; expensive
Uncertainty wt. (Kendall+ 2018)	Learn σ_k : $w_k \propto 1/(2\sigma_k^2)$	Extra parameters; can collapse
ReLoBRaLo (Bischof+ 2021)	Relative ratios + random lookback + smoothing	Lightweight; robust; no extra params to learn

ReLoBRaLo's appeal: no gradient-norm computation, no extra learnable parameters, strong empirical performance across PINNs, DEQNs, and multi-task regression.

ReLoBRaLo — Relative Loss Balancing with Random Lookback

Key innovations (Bischof et al., 2021):

1. **Relative balancing:** Use *ratios* $\mathcal{L}_i^{(t)} / \mathcal{L}_i^{(t-1)}$ instead of absolute values
2. **Random lookback:** With probability ρ , compare to **initial losses** $\mathcal{L}_i^{(0)}$ instead of the previous step
3. **Exponential smoothing:** Blend new weights with history via α

Why random lookback?

- ▶ Prevents weight drift
- ▶ “Resets” to global perspective
- ▶ Stabilises training
- ▶ Like simulated annealing for weights

ReLoBRaLo — Algorithm

Weight update at epoch t

- 1: Compute losses $\mathcal{L}_1^{(t)}, \dots, \mathcal{L}_K^{(t)}$
- 2: **Step-wise weights:** $\hat{w}_i^{(t)} = \text{softmax}_i\left(\frac{\mathcal{L}_i^{(t)}}{T \cdot \mathcal{L}_i^{(t-1)}}\right) \cdot K$
- 3: **Baseline weights:** $\hat{w}_i^{(0)} = \text{softmax}_i\left(\frac{\mathcal{L}_i^{(t)}}{T \cdot \mathcal{L}_i^{(0)}}\right) \cdot K$
- 4: **Combine with random lookback:**

$$w_i^{(t)} = \alpha \left[\rho w_i^{(t-1)} + (1 - \rho) \hat{w}_i^{(0)} \right] + (1 - \alpha) \hat{w}_i^{(t)}$$

- ▶ α : smoothing — how much to trust the historical weights
- ▶ ρ : lookback probability — how often to reset to initial comparison
- ▶ T : temperature — sharpness of the softmax

ReLoBRaLo — Experiment Results

	Equal weights	Inverse-loss	ReLoBRaLo
f_A mean error	5.2×10^{-1}	8.1×10^{-2}	3.4×10^{-3}
f_B mean error	3.7×10^0	1.5×10^0	8.2×10^{-1}
f_C mean error	4.5×10^1	9.8×10^1	5.1×10^1
Total rel. error	1.00	0.41	0.12

- **Equal:** f_C dominates training; f_A is essentially **not learned**
- **Inverse-loss:** Improves f_A , but introduces oscillations
- **ReLoBRaLo:** All three functions well approximated; $\sim 8\times$ lower total error

Loss Balancing for DEQNs

DSGE / Deep equilibrium nets:

- ▶ Euler equations at different scales
- ▶ Budget constraint residuals
- ▶ Market clearing conditions
- ▶ Each may have different units and magnitudes

⇒ ReLoBRaLo ensures all equilibrium conditions are learned, not just the largest.

ReLoBRaLo defaults:

Symbol	Name	Default
T	Temperature	1.0
α	Smoothing	0.999
ρ	Lookback prob.	0.999

Practical guidance

The defaults work well in most cases.
Only T needs occasional tuning.

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5. **Scaling Up — The IRBC Model**
6. Deep Surrogate Models
7. Hands-on Exercises

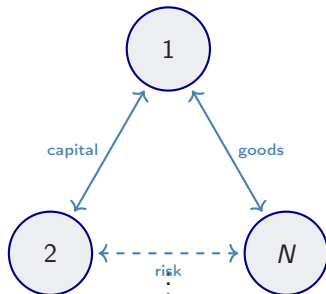
Why International Models?

Key questions in international macro:

- ▶ How do **capital flows** respond to productivity shocks?
- ▶ What governs **international risk sharing**?
- ▶ How do **trade imbalances** emerge?

⇒ Need models with N heterogeneous countries.

⇒ State space: $2N$ dimensions (capital + TFP per country).



Complete markets

IRBC Model Overview

International Real Business Cycle (IRBC) model:

- ▶ N countries, each with **country-specific capital** k^j and **TFP** z^j
- ▶ **Complete markets**: a social planner allocates resources with Pareto weights τ^j
- ▶ **Irreversible investment**: $I^j \geq 0$ (cannot disinvest)
- ▶ **Capital adjustment costs**: $\Gamma^j = \frac{\kappa}{2} k^j \left(\frac{k^{j'}}{k^j} - 1 \right)^2$

Dimensionality

State: $\mathbf{s} = (k^1, \dots, k^N, z^1, \dots, z^N) \in \mathbb{R}^{2N}$

Policy: $(k^{1'}, \dots, k^{N'}, \lambda, \mu^1, \dots, \mu^N) \in \mathbb{R}^{2N+1}$

Preferences and Production

CRRRA utility for each country j :

$$u^j(c) = \frac{c^{1-1/\gamma_j}}{1-1/\gamma_j}$$

- ▶ γ_j : intertemporal elasticity of substitution
- ▶ Heterogeneous: $\gamma_j \in [\gamma_{\min}, \gamma_{\max}]$

Social planner maximizes:

$$\max \sum_{t=0}^{\infty} \beta^t \mathbb{E} \left[\sum_{j=1}^N \tau^j u^j(c_t^j) \right]$$

Production in country j :

$$Y^j = A_{\text{tfp}} \cdot \exp(z^j) \cdot (k^j)^\zeta$$

TFP process (AR(1)):

$$z^{j'} = \rho_z z^j + \sigma_e (\varepsilon^j + \varepsilon^{\text{agg}})$$

$\beta = 0.99$	$\zeta = 0.36$
$\delta = 0.01$	$\kappa = 0.50$
$\rho_z = 0.95$	$\sigma_e = 0.01$

Equilibrium System

$2N + 1$ equations in $2N + 1$ unknowns:

- N Euler equations (relative error form):

$$\frac{\beta \mathbb{E}[\lambda' \cdot \text{MPK}^{j'} - (1 - \delta) \mu^{j'}] + \mu^j}{\lambda (1 + \Gamma_{k'})} - 1 = 0$$

- Aggregate resource constraint (1 equation):

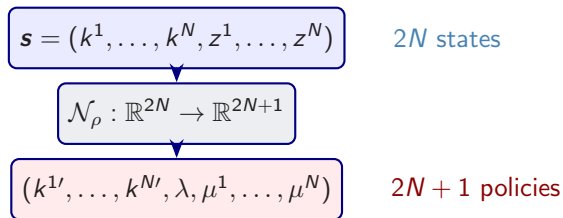
$$\sum_{j=1}^N \left[Y^j + (1 - \delta) k^j - k^{j'} - \Gamma^j - c^j \right] = 0$$

- N Fischer–Burmeister conditions (irreversibility):

$$\text{FB}(\mu^j, l^j) = \mu^j + l^j - \sqrt{(\mu^j)^2 + (l^j)^2 + \varepsilon} = 0$$

Consumption is determined by λ : $c^j = (\lambda / \tau^j)^{-\gamma_j}$.

DEQN Formulation for the IRBC



Loss: sum of squared residuals over all equilibrium conditions:

$$\ell_\rho = \frac{1}{B} \sum_{i=1}^B \left[\sum_{j=1}^N (\text{Euler}_j^{(i)})^2 + (\text{ARC}^{(i)})^2 + \sum_{j=1}^N (\text{FB}_j^{(i)})^2 \right]$$

Architecture: 2×64 hidden units, Swish activation, Softplus output (ensures positivity).

Scalability in N

N	States	Policies	NN params	Grid pts ($n=10$)
2	4	5	$\sim 4.7\text{K}$	10^4
4	8	9	$\sim 5.3\text{K}$	10^8
10	20	21	$\sim 6.7\text{K}$	10^{20}
50	100	101	$\sim 17\text{K}$	10^{100}
100	200	201	$\sim 30\text{K}$	10^{200}

- ▶ NN parameters grow **linearly** in N (only input/output layers scale)
- ▶ Grid methods grow **exponentially**: 10^{2N} points
- ▶ DEQN training cost per step: $\mathcal{O}(|\rho| \cdot B \cdot Q^{N+1})$
- ▶ For large N : replace tensor-product GH with monomial rules or QMC

From Brock–Mirman to IRBC

	Brock–Mirman	IRBC
Countries	1	N
States	(K, z)	$(k^1, \dots, k^N, z^1, \dots, z^N)$
Policies	C (or s)	$(k^{1'}, \dots, k^{N'}, \lambda, \mu^1, \dots, \mu^N)$
Equations	1 Euler	N Euler + 1 ARC + N FB
Constraints	none	irreversibility, adj. costs
Shocks	1	$N + 1$
Output act.	sigmoid	softplus
Analytical sol.	yes ($\delta = 1$)	no

The IRBC is a natural **scaling test** for the DEQN methodology.

IRBC Notebook

Hands-on: 04_IRBC_DEQN.ipynb

1. Parameters & steady state verification
2. Gauss–Hermite quadrature setup
3. Keras network definition
4. Model equations (production, consumption, adjustment costs)
5. Cost function with Euler, ARC, and FB residuals
6. Two-phase training loop
7. Convergence analysis & policy function plots
8. Economic diagnostics (consumption sharing, trade balance)

Self-contained: all equations implemented inline, no external imports beyond TensorFlow/NumPy.

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Motivation: Why Surrogates?

The computational bottleneck

Economic & financial models are expensive to evaluate:

- ▶ Solving Bellman equations / dynamic programming (DEQNs)
- ▶ PDE-based models (HJB, Black–Scholes for exotics)
- ▶ Monte Carlo simulations for risk measures

Key insight

We know the true model — we can generate *unlimited* training data by solving the model on a design of experiments.

Goal: Build a **surrogate** $\phi(\cdot | \theta_{NN})$ that approximates $f(\cdot)$ but evaluates *orders of magnitude faster*.

Look-up Tables $\rightarrow$ Neural Networks

Traditional: pre-compute & interpolate.

x	$\Phi(x)$
-2.0	0.0228
-1.0	0.1587
0.0	0.5000
1.0	0.8413
2.0	0.9772

Example: standard normal CDF table.

21st-century look-up table:

A neural network memorizes $\mathbf{x} \mapsto y$:

$$\phi(\mathbf{x}_t | \theta_{NN}) \approx y_t = f(\mathbf{x}_t)$$



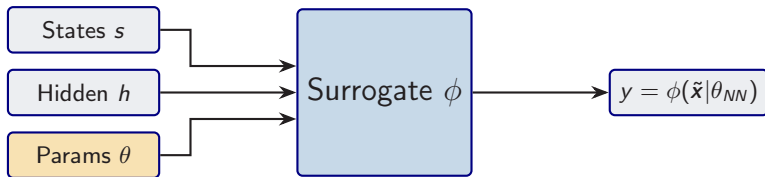
Continuous, differentiable, works in any dimension.

Pseudo-States: The Key Innovation

Core insight (Chen, Didisheim & Scheidegger 2026)

Treat model parameters θ as **extra state variables**:

$$\tilde{\mathbf{x}} = (\underbrace{s_1, \dots, s_n}_{\text{states}}, \underbrace{h_1, \dots, h_h}_{\text{hidden}}, \underbrace{\theta_1, \dots, \theta_p}_{\text{parameters}}) \in \mathbb{R}^d, \quad d = n + h + p.$$



Solve the model **once** over the full augmented space $\Rightarrow$ instant evaluation for *any* parameter configuration.

Building the Surrogate

1. **Design:** Draw $\tilde{\mathbf{x}}_i$ uniformly from $[\underline{x}, \bar{x}]^d$, $i = 1, \dots, m$.
2. **Label:** Compute $y_i = f(\tilde{\mathbf{x}}_i)$ using the expensive model.
3. **Train:** Minimize the empirical risk:

$$\hat{\theta}_{NN} = \arg \min_{\theta_{NN}} \frac{1}{m} \sum_{i=1}^m |\phi(\tilde{\mathbf{x}}_i | \theta_{NN}) - y_i|^2.$$

4. **Validate:** On a held-out test set, compute max / mean absolute error.

Remark

Since we can generate *unlimited* data from the true model, the training set can be made as large as needed $\Rightarrow$ **no overfitting risk** in the traditional sense.

Why Surrogates $\neq$ Standard ML

Standard supervised learning:

- ▶ Data is *scarce* and noisy
- ▶ Overfitting is a major concern
- ▶ Regularization, early stopping, cross-validation
- ▶ Small learning rates, careful tuning

Surrogate modelling:

- ▶ Data is **unlimited** (we own the model)
- ▶ Labels are **exact** (or noise-free)
- ▶ Can use large epochs, aggressive LR
- ▶ Focus on *approximation error*, not generalization

Once trained

- ▶ **Highly accurate** (errors $< 10^{-4}$ relative)
- ▶ **Orders of magnitude faster** than the original model
- ▶ **Differentiable**: gradients via automatic differentiation (backprop)

Speed Comparison: Option Pricing

From Chen, Didisheim & Scheidegger (2026), *J. Financial Economics*:

Task	FFT	Surr. (CPU)	Surr. (GPU)
Single price	23 ms	0.03 ms	—
1-day calib. (6×10^5)	3.3 h	18 s	0.4 s
1-year calib. (1.5×10^8)	≈ 100 d	1.2 h	98 s

Takeaway

Deep surrogates enable **real-time calibration** of option pricing models — tasks that previously took days or were infeasible.

Application I: Structural Estimation

Goal: Estimate structural parameters θ from data via GMM:

$$\hat{\theta}_{GMM} = \arg \min_{\theta} g_n(\theta)^\top W g_n(\theta),$$

where $g_n(\theta) = \frac{1}{n} \sum_{i=1}^n m(y_i, \theta)$ and computing $m(\cdot)$ requires solving the model.

Surrogate-based estimation

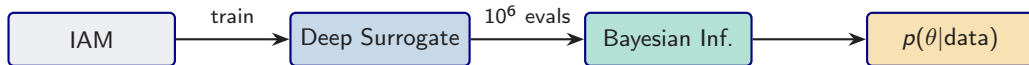
1. Replace $f(\cdot)$ with surrogate $\phi(\cdot \mid \theta_{NN})$.
2. **Automatic differentiation** gives analytical gradients $\nabla_{\theta} g_n$.
3. First-order conditions become a *system of equations* — solve in milliseconds.

Result: Full GMM estimation in **seconds** instead of weeks.

(Chen, Didisheim & Scheidegger 2026)

Application II: Uncertainty Quantification

DeepUQ (Friedl, Kübler, Scheidegger & Usui 2023):

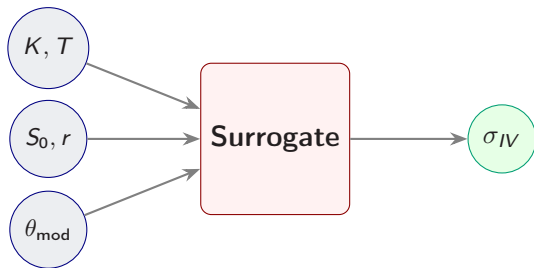


- ▶ Surrogate over 100+ climate & economic parameters
- ▶ Posterior distributions of damage functions and optimal carbon tax
- ▶ MCMC requires millions of model evaluations $\Rightarrow$ only feasible with surrogate

Application III: The Implied Volatility Surface

Goal: Learn the mapping from option characteristics to implied volatility:

$$\sigma_{IV} = \phi(K, T, S_0, r, \theta_{\text{model}})$$



Benefits:

- ▶ **Calibration:** Invert the surrogate to find θ_{mod} matching market prices.
- ▶ **Arbitrage-free:** Enforce constraints (no calendar/butterfly arb) in the loss.
- ▶ **Hands-on:** See `06_Surrogate_Primer.ipynb` for a Black–Scholes example.

Practical Tips for Surrogate Training

Architecture

- ▶ Swish activation ($x \cdot \sigma(x)$)
- ▶ 3–5 hidden layers, 128–512 neurons
- ▶ Linear output (no activation)

Training

- ▶ $m \geq 10^5$ samples (cheap to generate)
- ▶ Adam, LR 10^{-3} , reduce on plateau
- ▶ Hundreds of epochs (no overfit risk)

Validation

- ▶ Held-out test set
- ▶ Max / mean absolute error, R^2
- ▶ Scatter: prediction vs. truth

Loss functions

- ▶ MSE: $\frac{1}{m} \sum_i (\phi_i - y_i)^2$
- ▶ Relative L^1 : $\frac{1}{m} \sum_i |(\phi_i - y_i)/y_i|$

Gaussian Processes: A Complementary Tool

Definition

A **Gaussian Process** is a distribution over functions: $f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$.

Key properties:

- ▶ **Nonparametric**: model grows with data
- ▶ **Built-in UQ**: posterior mean *and* variance
- ▶ **Principled kernels**: encode smoothness, scale
- ▶ **Hyperparameter learning**: marginal likelihood

Posterior (noiseless):

$$\begin{aligned}\bar{f}_* &= \mathbf{k}_*^\top K^{-1} \mathbf{f} \\ \sigma_*^2 &= k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top K^{-1} \mathbf{k}_*\end{aligned}$$

Near data: **small variance**.

Far from data: **large variance** (reverts to prior).

Complementary to DNNs

GPs: best for $d \leq 10$ with built-in UQ. DNNs: best for $d \gg 10$ with large data.

GPs vs DNNs: When to Use Which

Criterion	Gaussian Processes	Deep Neural Networks
Uncertainty quantification	Built-in (posterior var.)	Requires ensembles
Hyperparameter tuning	Marginal likelihood	Grid / Bayesian opt.
Scaling with N	$O(N^3)$, limit $\sim 10^4$	$O(N)$ per epoch
Scaling with d	Curse of dimensionality	Handles $d \gg 100$
Small data ($N < 100$)	Excellent	Typically underfits
Interpretability	Kernel encodes prior	Black box

Rule of thumb

- ▶ $d \leq 10$, **need UQ**: Use GPs (+ BAL for efficient sampling).
- ▶ $d > 10$, **large data**: Use DNNs (deep surrogates).

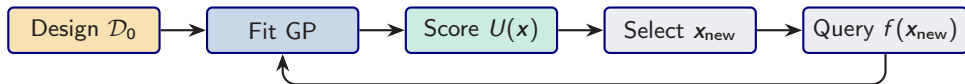
Bayesian Active Learning

Use the GP posterior variance to guide sampling.

Acquisition function

$$U(\mathbf{x}) = \rho \cdot \mu(\mathbf{x}) + \frac{\beta}{2} \log \sigma^2(\mathbf{x})$$

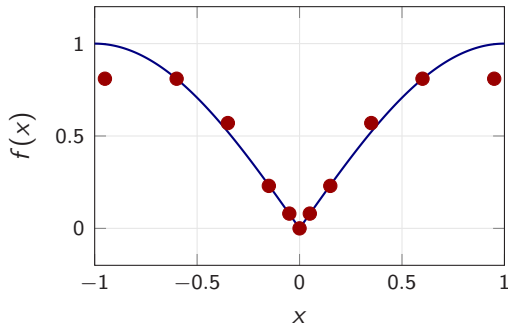
- ▶ $\mu(\mathbf{x})$: posterior mean (**exploitation**)
- ▶ $\sigma^2(\mathbf{x})$: posterior variance (**exploration**)
- ▶ Pure exploration ($\rho = 0$): select where uncertainty is highest



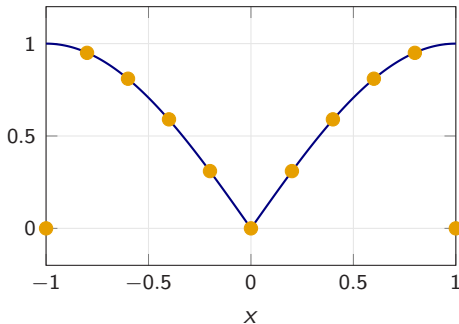
BAL vs Uniform Sampling

Target: $f(x) = |\sin(\pi x/2)|$ (has kinks at $x \in \{0, \pm 2\}$). Same budget: 11 points.

BAL: concentrates near kinks



Uniform: wastes points in smooth regions



BAL clusters points near $x = 0$ (kink) where the function is hardest to approximate. **Uniform** spreads evenly — several points wasted in smooth regions ($|x| > 0.5$).

Summary: Deep Surrogate Models

1. **Surrogates** = cheap, accurate replicas of expensive models.
2. **Pseudo-states** (θ as extra inputs) unlock:
 - ▶ Structural estimation (GMM in seconds)
 - ▶ Uncertainty quantification (MCMC with surrogate)
 - ▶ Optimal policy design
3. **Key advantage**: unlimited training data $\Rightarrow$ no overfitting.
4. **Speed**: orders of magnitude faster than original model.
5. **GPs + BAL**: complementary for moderate d with built-in UQ.

Hands-on notebooks

06_Surrogate_Primer.ipynb — Black–Scholes surrogate with IV inversion.

07_GP_and_BAL.ipynb — GP regression from scratch and Bayesian Active Learning.

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Code Notebooks

Notebook	Description
01_Brock_Mirman_1972_DEQN	Deterministic Brock–Mirman model with analytical benchmark
02_Brock_Mirman_Uncertainty_DEQN	Stochastic Brock–Mirman with TFP shocks
03_DEQN_Exercises_Blancs	Exercise problems (blanks to fill in)
03_DEQN_Exercises_Solutions	Exercise solutions
04_IRBC_DEQN	Full IRBC model: multi-country DEQN with adjustment costs and irreversibility
05_Neural_Architecture_Search	NAS via random search: hyperparameter space, top- k selection
06_Surrogate_Primer	Deep surrogates: Black–Scholes pricing & implied volatility inversion
07_GP_and_BAI	GP regression from scratch scikit-learn Bayesian Ac-

Suggested Readings

1. **Azinovic, Gaegauf & Scheidegger (2022).**
Deep Equilibrium Nets. International Economic Review 63(4), 1471–1525.
2. **Fernández-Villaverde, Nuño & Perla (2024).**
Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning. NBER Working Paper 33117.
3. **Chen, Didisheim & Scheidegger (2026).**
Deep Surrogates for Finance: With an Application to Option Pricing. Journal of Financial Economics 177, 104222.
4. **Friedl, Kübler, Scheidegger & Usui (2023).**
Deep Uncertainty Quantification: With an Application to Integrated Assessment Models. Working paper.

All papers are in the readings/ directory.

Questions?

Galo Nuño

Banco de España

Simon Scheidegger

HEC, University of Lausanne

<https://www.galonuno.com/>

[https:](https://sites.google.com/site/simonscheidegger/)

[//sites.google.com/site/simonscheidegger/](https://sites.google.com/site/simonscheidegger/)

Materials: [https://github.com/](https://github.com/sischei/DEQN_Galo_03_2026)

[sischei/DEQN_Galo_03_2026](https://github.com/sischei/DEQN_Galo_03_2026)

Key references:

Azinovic, Gaegauf & Scheidegger (2022)

Deep Equilibrium Nets

International Economic Review